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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Final Project Report
Section: C1 Group: 05

Construction of a 240V/260V Step Up Auto Transformer

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Abstract

The project aims to design, build, and test a 240V/260V step-up autotransformer with four tapping. It is designed to convert a primary voltage of 240V to secondary voltages of 260V, 220V, 180V, and 110V.

The autotransformer was developed to meet essential electrical engineering standards for reliability, and safety. The implementation phase involved real construction, testing, and data collection, followed by an in-depth evaluation of the design's efficacy. The autotransformer met all design criteria, with output voltages deviating from expected values by no more than 1%.

This project serves as a good reference for building similar step-up autotransformers, shining light on crucial design issues, material selection, and assembly processes.

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3 Introduction

We often face the problem of efficiently adjusting voltage levels in electrical systems, making them an essential component in power distribution and various industrial applications.

The problem in electrical power systems is complex because electrical loads are not constant; they fluctuate throughout the day and across seasons. The voltage levels must adapt to meet varying demand while avoiding overvoltage or undervoltage conditions that could damage equipment or disrupt power supply. Also electricity often travels over long distances from power generation sources to end-users. Voltage needs to be adjusted to minimize losses during transmission, as higher voltage reduces current, which, in turn, decreases resistive losses in the transmission lines. Again converting voltage levels in transmission lines involves much energy losses due to transformer losses and resistive losses

This problem can be solved by using an autotransformer. An autotransformer is a type of electrical transformer with only one winding. It's a transformer type that has a single winding that may be tapped at several locations to provide output voltage that is variable.
The several taps that autotransformer feature throughout their windings let us choose between various voltage ratios. The voltage adjustment will depend on the tap choice. In electrical systems, they are frequently employed to efficiently regulate voltage levels. As autotransformers don't need two independent windings like conventional transformers do, voltage levels can be adjusted effectively. The tap position of the autotransformer must be carefully chosen, and the output voltage must be regularly checked to make sure it stays within the appropriate range. An automatic transformer performs more efficiently than a transformer with two windings. Reduced transformer material results in lower core and ohmic losses. Because of the decreased voltage drop in the single winding's resistance and reactance, an auto transformer has superior voltage control.
Since autotransformers have some advantages over standard transformers, they are frequently utilized in the power supply and distribution industries. They have the benefit of being more compact, lighter, and less expensive than standard dual-winding transformers. Usually, it is convenient to incorporate an autotransformer in the system when a smaller stepping-up or stepping-down of the AC voltage level is necessary. Certain factors must be considered when building an autotransformer with a specified rating.

4 Design

4.1 Problem Formulation

4.1.1 Identification of Scope

The scope of this project is focused on the design, construction, and testing of an autotransformer with four tapings. The primary aim is to create a reliable and efficient autotransformer with a primary voltage of 240V and secondary voltages of 260V, 220V, 180V, and 110V. The project covers the following specific areas:

Design Phase:

1. Literature Review: Study of existing transformer designs, principles, and materials.
2. Calculations: Detailed calculations for determining the number of turns, wire gauge, and bobbin size.
3. Component Selection: Identification and selection of materials like the core, wire, and insulation.

Construction Phase:

1. Winding: Implementation of the winding process based on the design calculations.
2. Tapping: Creation of tapping at specific points on the winding to achieve the desired output voltages.
3. Insulation and Assembly: Ensuring proper insulation between turns and final assembly of the transformer.

Testing Phase:

1. Functional Testing: Verification of the output voltages and the open circuit test.

Documentation:

1. Project Report: Compilation of all phases of the project, including design, construction, and testing.
2. User Manual: Guidelines for safe and effective usage of the constructed autotransformer.

Limitations:

1. Scale: The project is limited to a primary voltage of 240V and is not intended for industrial-scale applications.
2. Material Constraints: The project uses only readily available materials.

By clearly defining this scope, the project aims to maintain focus and deliver a high-quality, reliable, and safe autotransformer within the constraints of time, budget, and resources.

4.1.2 Literature Review

Introduction

The literature review aims to provide an overview of existing research, theories, and practices related to the design and construction of autotransformers. The focus is on understanding the principles behind autotransformers, their applications, and the key considerations in their design and construction.

Basic Principles of Autotransformers

An autotransformer uses a single winding for both the primary and secondary circuits and operates on

the basis of electromagnetic induction. It is more compact and efficient than typical transformers. The voltage transformation is accomplished via tappings at various places along the winding that correspond to the desired output voltages. However, because it lacks electrical isolation between primary and secondary circuits, it is unsuitable for applications requiring high voltages.

Applications of Autotransformers

Autotransformers are commonly used in applications where voltage needs to be stepped up or down but where electrical isolation between the primary and secondary is not critical. They are often found in audio systems, railway traction, and power distribution networks.

Design Considerations

1. Voltage and Ratings: The voltage and power ratings are fundamental parameters that dictate the design of the autotransformer.
2. Turns Ratio: The turns ratio is determined by the voltage ratio between the primary and secondary windings and is crucial for the design.
3. Wire Gauge: The wire gauge must be selected based on the maximum current that the winding will carry, which is determined by the power rating and voltage.
4. Core Material: The choice of core material affects the efficiency and performance of the transformer. Common materials include iron, ferrite, and amorphous metal.
5. Insulation: Proper insulation is essential for the safety and longevity of the transformer.

Previous Works

Several research papers and patents have been published on the optimization of autotransformer design. Studies have explored the use of different core materials, winding techniques, and cooling methods to improve efficiency and performance.

1. Scilab Textbook Companion for Electrical Machine Design by A. K. Sawhney
2. Electrical Machine Design by V. Rajini and V. S. Nagarajan
3. A Course in Electrical Machine Design by A. K. Sawhnby

4.1.3 Formulation of Problem

This project addresses a major challenge in modern industrial operations which is designing and constructing an improved auto transformer to enable precise voltage management as well as greater energy efficiency. Voltage stability is essential in industrial processes, preventing costly equipment damage and ensuring consistent output levels.

Long power lines suffer from voltage drop in high-current applications. To compensate for this voltage loss, autotransformers might be utilized.Besides the inefficiencies of traditional auto transformers contribute to higher energy bills. Therefore, our project takes on the challenge of designing an auto transformer that not only stabilizes voltage but also significantly reduces energy consumption, and increases its efficiency as because autotransformer has both conduction power and induced power ,for which autotransformer has greater power flow than two winding transformer .

So, The major goal of this project is to design and build a step-up autotransformer capable of delivering four different secondary voltages (260, 220, 180, and 110) from a single primary voltage of 240V. The transformer should be efficient, dependable, and safe to use.

4.1.4 Analysis

The project's efficiency hinges on its ability to transform a single primary voltage of 240V into four distinct secondary voltages (260, 220, 180, and 110V) with minimal energy losses. Efficiency gains will be realized through optimal core material selection depending on its availability, wire gauge

(SWG) choices, and winding design. Rigorous testing and data analysis will quantitatively measure efficiency improvements compared to conventional transformers. Voltage stability and regulation are equally vital, with advanced control algorithms and tap-changing mechanisms ensuring voltage remains within a minimum nominal value under varying loads. The project also prioritizes environmental impact, reduce energy consumption, high efficiency.

4.2 Design Method

Formula and Calculations for windings of auto transformer:

An autotransformer has a single winding with two end terminals and one or more terminals at intermediate tap points. It is a transformer in which the primary and secondary coils have part of their turns in common. The portion of the winding shared by both the primary and secondary is the common section. An autotransformer can act either as step-up or step-down transformer depending on which terminal (HT or LT) is connected with input AC voltage. The following two figures illustrate this:

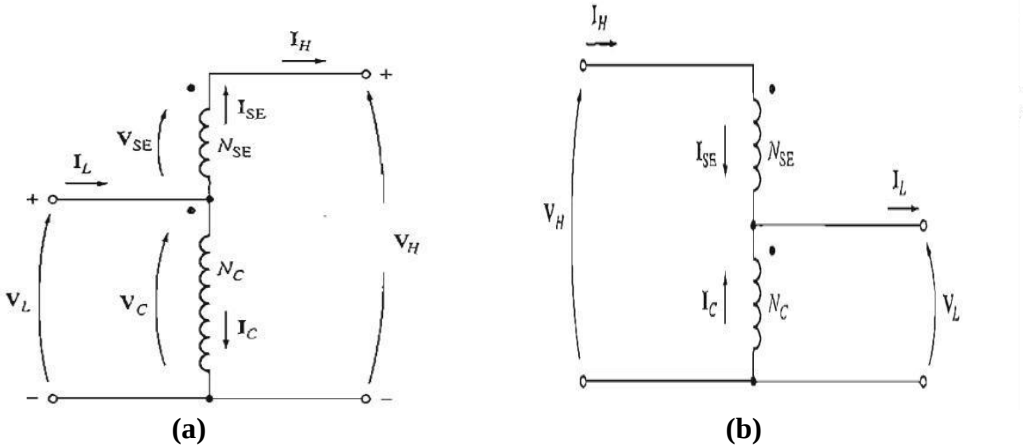


Figure: (a) Step-up Autotransformer ,(b)Step-down Autotransformer

V_L = Voltage on low voltage side

V_H = Voltage on high voltage side

I_L =Current on low voltage side

I_H = Voltage on low voltage side

Now, $V_H = V_C + V_{SE}$ ……(i)

But, $\frac{Vc}{V_{SE}} = \frac{Nc}{N_{SE}}$

and $V_C=V_L$, so

$V_H = V_L + \frac{N_{SE}}{Nc} V_L$

Or, $\frac{V_L}{V_H} = \frac{Nc}{Nc+N_{SE}}$ … … … (ii)

The current relationship between the two sides of the transformer can be found .

$I_L=I_C+I_{SE}$……(iii)

As $I_C=\frac{N_{SE}}{N_c} I_{SE}$ & $I_{SE}= I_H$,

therefore-

$I_L = \frac{N_{SE}}{N_c} I_H + I_H$

$\frac{I_L}{I_H} = \frac{Nc+N_{SE}}{Nc} \dots\dots\dots (iv)$

The Apparent Power Rating Advantage of Autotransformers:

The input and output apparent power of a transformer are equal.

That is , $S_{in}=S_{out}= S_{IO}$ …….(v)

Here, S_{IO} = input and output apparent power of autotransformer,

$S_{in}=V_LI_L$ & $S_{out}=V_HI_H$.

The apparent power in the winding of the transformer is-

$S_W=V_CI_C$

$=V_L(I_L-I_H)$

$= V_LI_L- V_LI_H$

$= V_LI_L - V_LI_L\frac{Nc}{Nc+N_{SE}}$

$=V_LI_L\frac{N_{SE}}{Nc+N_{SE}}$

$= S_{IO}\frac{N_{SE}}{Nc+N_{SE}}$……..(vi)

We get ,$\frac{S_{IO}}{S_w} = \frac{Nc+N_{SE}}{N_{SE}}$……(v)

Equation(v) describes the apparent power rating advantage of an auto-transformer over a conventional transformer.
For designing an auto transformer following equipment are needed:

Equipment:
1. Core:

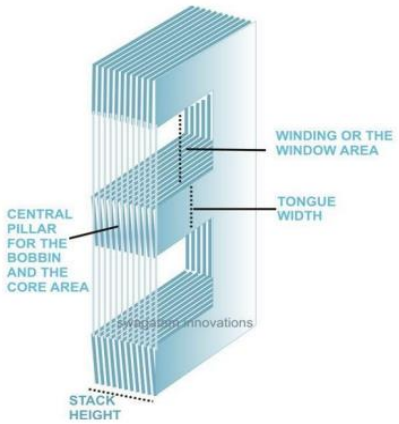


Figure: Core

Here the core area = tongue width*stack height,
The relationships among core area, tongue width, turns per volt and power consumptions are :
Turns per volt is inversely proportional to area ,where as power is proportional to area. So if tongue width is reduced ,area will be also reduced ,turns per volt will be increased as a result winding area as well as wire length will be increased.

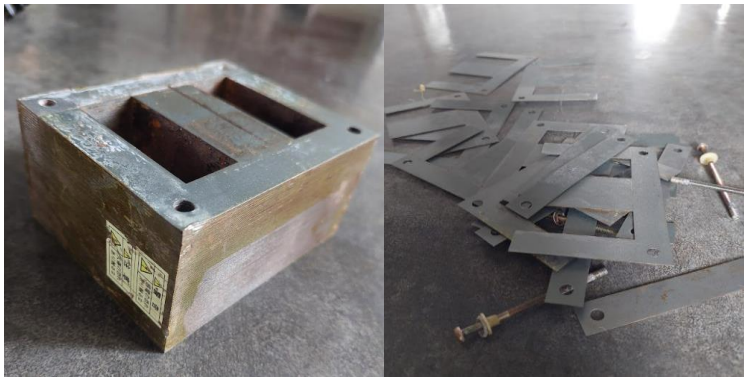


Figure:EI core-4

So, bringing all these things in optimization for designing the core, we have chosen EI core-4.

2. Bobbin:

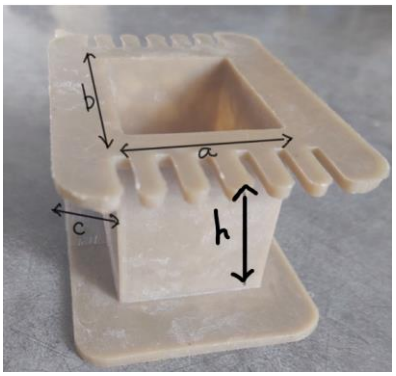


Figure: Bobbin

b = stack height
a = tongue width
h = Tongue length
c= Window width
window area = h*c

3. Wire:

From standard wire gauge (SWG) chart:

Max Current Capacity (Amp)	**Turns/Sq.cm**	**SWG**
1.622	87.4	19
1.313	106	20
1.0822	130	21
0.8945	176	22
0.5838	242	23

Winding area needed to be considered as to fit the winding within the window area.Also it has to be kept in mind that current flowing through wire doesn't exceed the rated current.
We choose **Gauge 20** initially.



Figure :Wire

4. Tapping tube



Figure : tapping tube

5. Insulation paper

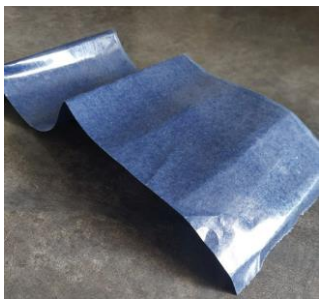


Figure: Insulation paper

6. Connector

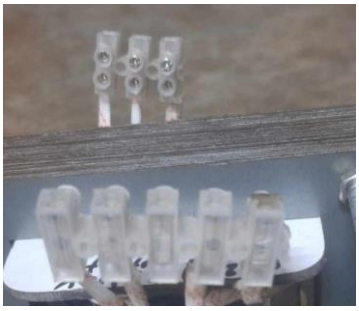


Figure: Connector

Formula & Calculations for core, bobbin and wire selections:

Core _ Area, A=$1.152\sqrt{Sw}$

Turns per volt = $\frac{1}{4.44\times10^{-4}\times f\times A\times B}$

Here, f=frequency of the power supply

B= flux density. (We are going to use CRGO-grade laminated sheet steel and maximum flux density for this material is approximated)

common winding current, =$S_{IO}[\frac{1}{N_C}-\frac{1}{N_C+N_{SE}}]\frac{1}{\eta\%}$

Here η =efficiency.

Total Window Area of core:

Input winding area, A_I=$\frac{Input\ winding\ turns}{Turns/Sq.cm}$

Output winding area,A_O=$\frac{output\ winding\ turns}{Turns/Sq.cm}$

A_{Ins} =Insulating area

Total Window Area, A_T = (A_I + A_O + A_{In})*1.3

Gross area, A_G =1.1×(Calculated core area)

Tongue width, T_W = $\sqrt{A_G}$

Calculations:

At first , lets design for 200VA auto-transformer . The optimum cost effective design for 200VA auto-transformer would be:

S_W= $S_{IO}\frac{N_{SE}}{Nc+N_{SE}}=200\times\frac{260-240}{260}=15.385$ VA.

Core _ Area ,A=$1.152\sqrt{Sw}=1.152\sqrt{15.385}=4.518\ sq.cm$.

Turns per volt = $\frac{1}{4.44\times10^{-4}\times f\times A\times B}=\frac{1}{4.44\times10^{-4}\times50\times4.518\times1.3}=7.67$

Common winding current, =$200[\frac{1}{240}-\frac{1}{260}]\frac{1}{92\%}$=0.0696 A

Here let's assume η =92% .

Output winding :

Output current =$\frac{200}{260}=0.769A$

Total Window Area of core:

Input winding area, A_I = $\frac{Input\ winding\ turns}{Turns/Sq.cm}$=$\frac{240\times7.67}{106}$=17.366 cm^2

Output winding area, A_O = $\frac{output\ winding\ turns}{Turns/Sq.cm}$=$\frac{(260-240)\times7.67}{106}$=1.447 cm^2

A_{Ins} =Insulating area

Total Window Area, A_T = (A_I + A_O+A_{Ins})*1.3 = 1.3*(17.366+1.447)= 24.46 cm^2

Gross area ,A_G =1.1×(Calculated core area)=$1.1\times4.518=4.97$ cm^2

Tongue width, $T_W = \sqrt{A_G} = \sqrt{4.97}$=2.23 cm

Iterative Design Process:

Now, our objective is to:
- Scout for readily accessible cores that closely match the calculated Tongue-Width and Window Area values. Our initial investigation indicates that the E-I 32 core is suitable for our transformer.
- In the event of unavailability, consider increasing the core area.
- It's important to note that increasing the core area leads to a reduction in the turns/voltage value, which in turn significantly diminishes the window area.
- In our country, most cores are primarily designed for 2-winding transformers, resulting in smaller window areas.
- This iterative process is repeated multiple times until we arrive at the most optimal parameters, taking into account both availability and economic advantages.

After performing iterations we came to the following conclusions:

Parameters	**Values**
Wire Gauge (SWG)	22 SWG
Total Window Area (Needed)	9.47cm^2
Total Window Area (Available in E-I 4 core)	12.5 cm^2
Tongue Width	3.3cm
Core Number	EI-4
Stack Height	2.7 cm
Bobbin's Width	5.5cm

Core _ area = 3.3× 2.7 = 8.91cm^2
The corresponding Core type that suits is , **CRGO E-I 4(Shell type iron core stamping).**
Turns per volt $= \frac{1}{4.44\times10^{-4}\times f\times A\times B} = \frac{1}{4.44\times10^{-4}\times50\times8.91\times1.3} = 3.89$

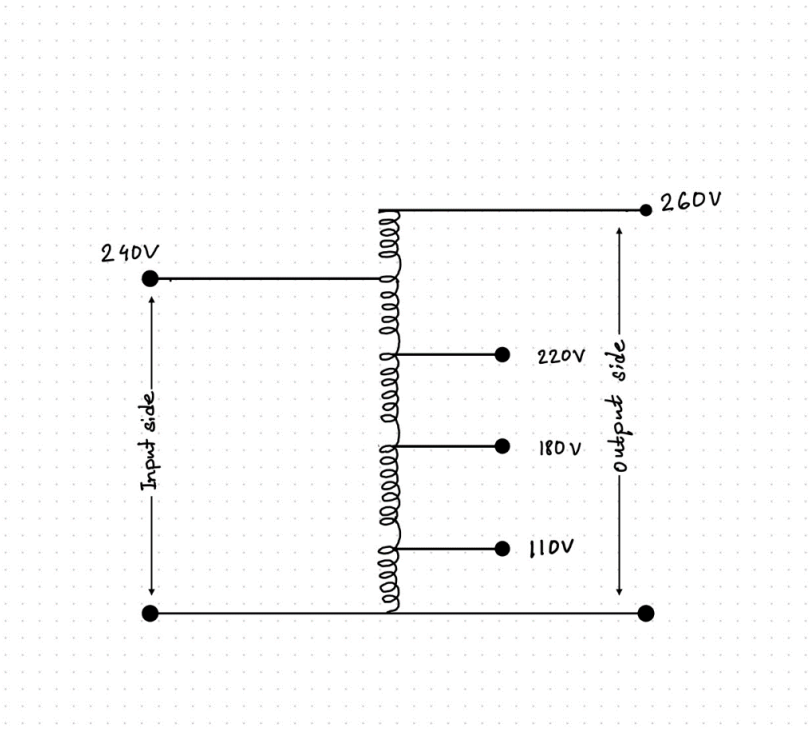
N.B : Finally to induce 1 volt without saturation we need approximately 3.89

Tapping :

Voltage	Turns
260	1012
240	933
220	855
180	700
110	428

Hence total turns = 1012

4.3 Circuit Diagram



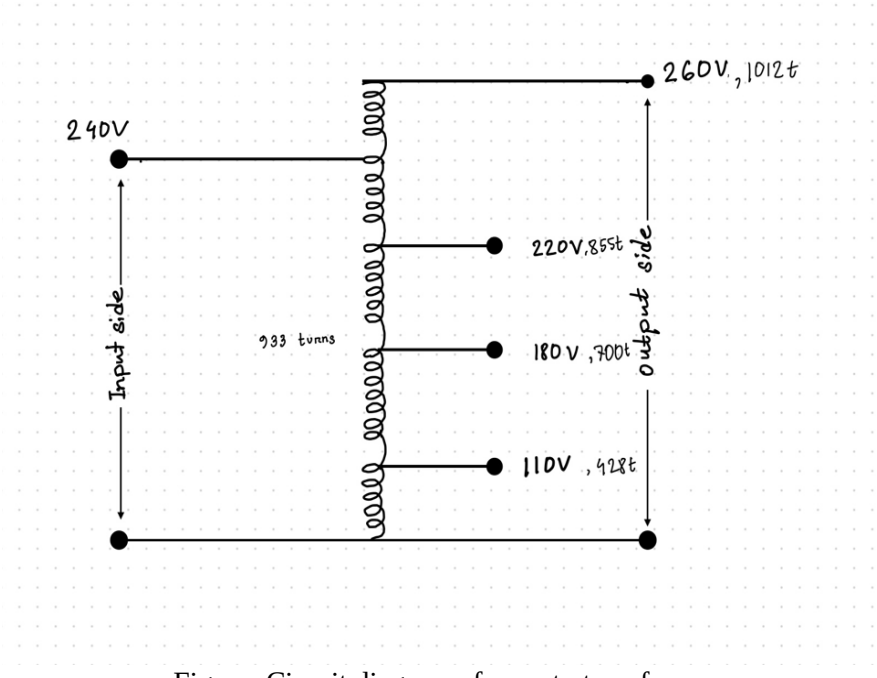


Figure :Circuit diagram of an auto transformer

5 Implementation

5.1 Description

Description:
Through following steps we constructed the auto transformer;
1. Custom Bobbin Building: We custom-built the bobbin according to optimized size parameters. This step ensures that the bobbin perfectly accommodates the winding and cores, optimizing the transformer's performance.

2.Wire Selection and Winding: We used 22 SWG wire and winding apparatus for the winding process. Achieving 1012 turns, including taps at appropriate places, is a crucial step to meet the project's voltage transformation requirements.

3. Insulation: To ensure safety and reliability, insulation paper was used to cover the exposed winding. This step is critical to prevent electrical short circuits and protect against potential damage.

4.Core Installation: The E-I cores were installed one by one. Proper core installation is essential to maximize magnetic coupling and efficiency.

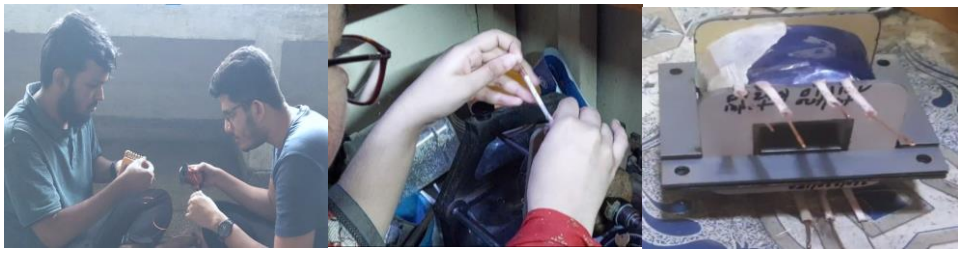


Figure :winding ,tapping,core installation

.
Our implemented 240V/260V step up auto transformer's construction and specifications are given below :

Parameters	**Values**
Wire Gauge (SWG)	22 SWG
Total Window Area (Needed)	9.47 cm^2
Total Window Area (available)	12.5 cm^2
Tongue Width	3.3 cm
Core Number	EI-4
Stack Height	2.7 cm

Voltage rating :

Voltage on input side	turns	Voltage on output side	Turns
240	933	260	1012
		220	855
		180	700
		110	428




Figure : auto-transformer

5.2 Experiment and Data Collection

After constructing the transformer ,open circuit test was performed in the laboratory and following data has been obtained

Voltage rating :

Voltage on input side	turns	Voltage on output side	Turns
240	933	261	1012
		221	855
		181	700
		110	428

Open circuit test (LT):

Open circuit voltage,V_{oc}	Open circuit current,I_{oc}	Power,P_{oc}
240 V	0.095 A	3 W



Figure: Data testing

N.B: We cannot perform a short-circuit test on an auto-transformer , because it's a single winding transformer .Short-circuit test on an auto-transformer is nothing but shorting the power supply completely .Rather we can perform partial loading test.

5.3 Data Analysis

Determination of error in output side:

Voltage on input side	Voltage on output side		|%Error|
	theoretical	experimental	
240	260	260	0.38%
	220	221	0.45%
	180	181	0.55%
	110	110	0.00%
Turns per Volt	1012/260=3.89	1012/261=3.87	0.51%

Determination of R_c and X_m :
Using the values of Open-circuit test we can determine the values of RC and XM of the magnetizing branch.

1. Power factor ,$pf = \frac{P_{OC}}{V_{OC}\, I_{OC}} = \frac{3}{240\times 0.095} = 0.136$

Now ,excitation admittance, $Y_E = \frac{I_{OC}}{V_{OC}} \angle -\cos^{-1} pf$

$$= \frac{0.095}{240} \angle -\cos^{-1}(0.136)$$
$$= 5.38 \times 10^{-5} - j3.92 \times 10^{-4}$$
$$= \frac{1}{R_C} + \frac{1}{jX_M}$$

So,Therefore ,the values of the excitation branch referred to LT side are
$R_C = \frac{1}{5.38\times 10^{-5}} = 18587.36\ \Omega$,
$X_M = \frac{1}{3.92\times 10^{-4}} = 2551.02\ \Omega$

5.4 Results

So from analyzing the data we found,

- The errors of the voltages of secondary side are almost negligible. From which we can say that our theoretical calculations for choosing proper core size ,bobbin size ,dimension, swg of wire were almost correct.
- Output voltage is 261 V instead of 260 V ,this is because some extra winding might be done unintentionally .Our Turns/Voltage was ≈ 3.89 ,and error is 0.5%,which is negligible.
- The one tap gave an accurate voltage output (110V).
- The finding values of Rc =18587.36 Ω, and X_M =2551.02 Ω .

5.5 YouTube Link

https://youtu.be/cXBzgdiIiOc?si=voRHemtnCxtDrI2t

6 Design Analysis and Evaluation:

6.1 Novelty

Our project was centered around designing and constructing a 240/260V step-up autotransformer with

four tapping. A particular use case for such a transformer might be in smaller-scale applications, such as home or educational contexts, which is in contrast to many transformer projects that aim for large-scale, industrial applications. Although the project follows the fundamentals of electrical engineering, it stands out by showcasing how a very straightforward, low-power transformer can be created, tested, and designed with great accuracy and dependability. The tapping have been designed to give relevant secondary voltages, that can be used for a variety of applications in our regular day to day life. The compactness improves thermal performance, which results in improved efficiency and lower operating costs. The project offers a thorough manual that might be used to other low-power applications, making it a useful educational model. This makes it a distinctive contribution to the discipline, especially helpful for those seeking an easier, more accessible way to comprehend the fundamentals of transformer design and testing.

6.2 Design Considerations

The design of the 240/260V step-up autotransformer with tappings at 260V, 220V, 180V, and 110V was influenced by several important considerations. These considerations were not only technical but also extended to public health and safety, environmental impact, and cultural and societal needs.

6.2.1 Considerations to Public Health and Safety

- Electrical Safety: The autotransformer was designed with proper insulation and grounding to minimize the risk of electrical shocks. All components were selected based on their ability to meet or exceed safety standards.
- User Guidelines: Clear and concise user guidelines were developed to ensure that end-users can operate the autotransformer safely, thereby reducing the risk of accidents due to misuse.

6.2.2 Considerations to Environment

The design aims to minimize energy loss, thereby reducing the transformer's carbon footprint over its operational lifetime.

6.2.3 Considerations to Cultural and Societal Needs

- Accessibility: The design aims to be user-friendly and accessible to people with varying levels of technical expertise, making it suitable for educational as well as domestic settings.
- Affordability: Cost-effective materials and construction methods were chosen to make the autotransformer affordable, thereby increasing its accessibility to a wider range of the population.
- Educational Value: The autotransformer serves not just as a functional device but also as an educational tool, providing insights into electrical engineering principles, which can be valuable in academic settings.

  By taking these considerations into account, the design aims to be not only technically sound but also socially responsible and environmentally sustainable.

6.3 Limitations of Tools

- The rated apparent power is 200VA. We cannot exceed it for safety reasons
- We can use it only in the limited places where a slight variation of the output voltage from input voltage is required.
- The primary winding and the secondary are directly connected. As a result, the input and output are no longer completely separated.

- Compared to the two-winding transformer with the same rating, the short-circuit current is significantly greater.

6.4 Impact Assessment

6.4.1 Assessment of Societal and Cultural Issues

The autotransformer can grow a lot of societal issues if it is used in different fields of machineries. The societal and cultural issues can be-

1. **Cultural device improvement:** With the concept of autotransformer ,we can make the machines using art works, more easier to work with (even without the artists behind them) but this artificial transform will make our cultural pride deemed.
   Counter logic: Culture is not an immortal thing. With flow of time culture becomes replaced, reformed and re-innovated. May linear autotransformer can have a negative change there, but if there were autotransformers ,this negative changes will come also in other forms with the flow of time.
2. **Profession loss:** In case of making use of autotransformer in different devices will make it easier to work with but we may have to lose the professionals who work behind all the common devices .We may have to replace them with engineers and technicians.
   Counter logic: New technology all time degrades the present existence of professional systems but it also opens doors for those who can cope with the newer situations and improvements. So the professionals will be changed, profession will be changed but the working rate or worker number will not die out.
3. **Prohibition against cultural or societal independence:** In recent days, we have encountered some problems regarding the metro-rail caused by kite flying, which is nothing less than a symbol of a citizen's cultural freedom but due to interference in the power system of Metrorail we had to make cross amendments in corresponding sections of rules. The same is going to happen when we will have the intention to use autotransformer in different mechanisms. We have to bring new laws that will restrict some of the freedom experienced by a common citizen in the present condition.
   Counter logic: Laws and technologies are for the betterment of people. Maybe autotransformer will introduce some newer restrictions but the impact of using autotransformer is so much helpful that the impact of these restrictions are negligible with respect to the impact of the introduction of autotransformer technology

   .

6.4.2 Assessment of Health and Safety Issues

Some safety and health issues are there regarding linear induction motors like

1. **Electrical safety issues:** As high voltage will be used in autotransformer, a lot of chances are there regarding electrical accidents. So safety issues are a great source of concern.
   Counter logic: The designing of a machine is all time done giving the highest priority to electrical safety. So as we designed our device to minimize electrical harm, may it can be used safely. To provide more safety, we can add additional devices in parallel with it like fuses and other control devices.
2. **Personal Protective Equipment (PPE):** Ensure that all workers have access to and are wearing the appropriate PPE for the tasks they are performing. This may include safety helmets, gloves, safety glasses, ear protection, and flame-resistant clothing.
3. **Equipment Inspection and Maintenance:**Regularly inspect and maintain all equipment used in the autotransformer tapping project. This includes transformers, switches, cables, and any associated tools. Any damaged or malfunctioning equipment should be repaired or replaced promptly.

6.4.3 Assessment of Legal Issues

Some legal issues that can be encountered while using linear induction motors like

1. **Linear induction motor placement:** As autotransformer will be lighter than a general average-weighted machine, we need to design such a civil engineered frame that will not face any harm due to at least richter 7.0 earthquake. But in our country some buidings are registered as safe for operation with this type of machine, but they do not show proper result regarding this.
   Counter logic: Before implanting any autotransformer an civil engineered inspection can be made mandatory to prevent the problems regarding it.
2. **Criminal issues:** As we can give the secondary a higher output than the primary by improving core , turn number ,rotor and wire,it can be used for any kind of automated killing or harming missions.
   Counter logic: Necessary laws can prevent people from implementing this kind of misdeeds.

6.5 Sustainability and Environmental Impact Evaluation

Sustainability evaluation: The linear autotransformers are very much sustainable if we take the rated measurements in concern while using the devices. Again, as there are not a lot of machineries in its construction, the system is easier to control and easier to keep it in its safe and life-enhancing working area.

Environmental Impact Evaluation: The autotransformer will have a very little environmental negative impact in fields like

1. Core production system.
2. Electricity production system

On the other hand ,directly autotranformers have no environmental bad impact. So it can be safely used in open environment.

6.6 Ethical Issued

No kinds of unfair means was adopted during the whole project. We went to the shop (Nawabpur & Dholaikhal) and bought all the required materials. Then Built the autotransformer from scratch. We actively did the winding and insulated it with insulation paper. All the wiring were also done by hand in a 4-nail structure. Every member of the group spontaneously participated in every steps from winding to tapping process.

Environmental Impact: Consider the potential environmental consequences of the project, such as soil contamination, air emissions, or habitat disruption. Ethical concerns may arise if the project does not incorporate environmentally responsible practices or fails to mitigate its ecological footprint.

Transparency and Disclosure: Maintain transparency in project planning, execution, and reporting. Ethical concerns may arise if information is withheld from stakeholders or if there is a lack of transparency in financial transactions.

Integrity in Reporting: Ensure that project progress and outcomes are reported accurately and honestly. Ethical concerns can arise if information is manipulated to present a false image of the project's status or performance.

Respect for Indigenous Rights: If the project impacts indigenous communities or territories, respect their rights, traditions, and cultural heritage. Ethical dilemmas can occur if these rights are not recognized or protected.

Long-Term Consequences: Consider the long-term consequences of the autotransformer tapping project, both positive and negative, and make ethical decisions that prioritize the well-being of future generations.

Ethical Leadership: Project leaders and decision-makers should demonstrate ethical leadership by setting a strong example, adhering to ethical principles, and promoting ethical behavior throughout the project.

Conducting ethical risk assessments, and incorporating ethical considerations into decision-making processes are essential steps in ensuring an ethically responsible project.

7 Reflection on Individual and Team work

7.1 Individual Contribution of Each Member

ID	Construction	Report	Presentation
2006157	Winding, tapping, core	Merging (page set up, arranging, content, figure), Investigations (PO(d)), Literature Review, Experiment Design, Data Analysis and Interpretation	Slide
2006158	Winding, tapping, core	Calculation, Design, Design Method, Experiment and Data Collection**,** Data Analysis**,** Circuit Diagram, Bill of Materials, Calculation of Per Unit Cost of Prototype, Calculation of Per Unit Cost of Mass-Produced Unit	Slide
2006159	Winding, tapping, core	Abstract, Design, Identification of Scope, Literature Review, Executive Summary, User Manual, References	Slide, Video Editing
2006160	Winding, tapping, core	Introduction, Design Analysis and Evaluation, Novelty, Limitations of Tools (PO(e)), Reflection on Individual and Team work (PO(i)), Individual Contribution of Each Member, Mode of Team Work, Diversity Statement of Team, Log Book of Project Implementation, Communication (PO(j)), Future Work (PO(l)), report merging	Slide, Capturing images and videos

2006161	Winding, tapping, core	Calculation, Youtube link, User Manual, Project Management and Cost Analysis (PO(k)), Bill of Materials, Calculation of Per Unit Cost of Prototype, Calculation of Per Unit Cost of Mass-Produced Unit, Timeline of Project Implementation	Slide
2006162	Winding, tapping, core	Design Analysis and Evaluation, Design Considerations (PO(c)), Impact Assessment (PO(f)), Sustainability and Environmental Impact Evaluation (PO(g)), Ethical Issued (PO(h))	Slide
2006163	Winding, tapping, core	Calculation, Design, Formulation of Problem, Analysis, Design Method, Implementation, Experiment and Data Collection, Data Analysis, results, Reflection on Individual and Team work (PO(i)), Individual Contribution of Each Member, Circuit Diagram, report merging.	Slide, Capturing images and videos



Figure :Contribution of team members

7.2 Mode of TeamWork

We know there are 4 modes of teamwork:

1. COULD DO - Sharing ideas and generating many different possible courses of action
   Timeline: 6th week
2. SHOULD DO - Learning sessions and retrospectives.
   Timeline: 7th week
3. WILL DO - Decision and action meetings.
   Timeline: 8th week
4. DOING - Actually making and creating stuff, and moving the work forward
   Timeline : 9th – 12th week

7.3 Diversity Statement of Team

There was not much diversity in the statements of the team members. All of us completed the project with mutual understanding.

7.4 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
02.07.23	Project Proposal	160 & 163: gave various suggestions about project 159 & 161: made slides	Had discussion about project	Zoom meeting
03.07.23	Assigned with this project	158 & 161: introduced everyone with various	Discussion	Studied autotransformer

		aspects of this project		
07.07.23	Official Discussion part 1	157: gave some ideas	All went through the ideas	Though how to implement these
11.07.23	Official Discussion part 2	163: gave alternatives 162: went through the ideas and gave feedback	Everyone gave feedback	Feedback discussion
17.07.23	Offline meeting 1	160: gave some possible design 158: showed the calculations 159: gave design implementation ideas	Did calculations	
25.07.23	Online meeting	157 & 161: gave a short plan 160 & 163: gave the sources about the project	A rough plan for the project	Disission making session
03.08.23	Final planning	157,159,162: gave the contacts regarding the shops for equipment	Discussed the dates for working	
11.08.23	Buying equipment	158 & 161: bought the equipment	Planned the budget	Work started
15.08.23	Offline working day 1	159: dismantled the core 160: winding 163 & 157: calculations	Started to make the autotransformer	
21.08.23	Working day 2	161,162 and 163: winding	All calculated the values	
25.08.23	Working day 3	Everyone finished winding tapping and core		Thought the work is done
01.09.23	Testing	157,158,159,160,163 went for testing		The results were not accurate
04.09.23	Working day 4	All reconstructed the project		
07.09.23	Testing	158,161 went for testing	All check the results	Finally finished

8 Communication

8.1 Executive Summary

This project successfully designs, constructs, and tests an autotransformer with four tappings, converting a 240V primary voltage to secondary voltages of 260V, 220V, 180V, and 110V.. The autotransformer meets all design specifications. Comprehensive testing confirms output voltages within 1% of expected values. The project also offers a practical guide for similar electrical engineering endeavors.

8.2 User Manual

Table of Contents

1. Introduction
2. Safety Precautions
3. Specifications
4. Installation Guide
5. Operating Instructions
6. Troubleshooting
7. Maintenance

1. Introduction

This manual provides essential information on how to install, operate, and maintain the autotransformer safely and efficiently.

2. Safety Precautions

1. Always disconnect power before installing or servicing the transformer.
2. Do not operate the transformer in wet or damp conditions.
3. Ensure proper grounding before use.
4. Keep away from children and pets.
5. Use only as directed in this manual.

3. Specifications

1. Primary Voltage: 240V
2. Secondary Voltages: 260V, 220V, 180V, and 110V
3. Frequency: 50 Hz
4. Wire Gauge: 22 SWG
5. VA rating : 200 VA

4. Installation Guide

1. Location: Choose a dry, well-ventilated location.
2. Grounding: Connect the ground wire to a proper earth ground.
3. Input Connection: Connect the primary wires to your 240V AC source.
4. Output Connection: Connect the secondary wires to your load, choosing the appropriate tapping for your voltage needs.

5. Operating Instructions

1. Power Up: Turn on the power supply to the primary winding.
2. Voltage Selection: Choose the appropriate tapping for your required secondary voltage.
3. Load Connection: Connect the load to the selected secondary tapping.

6. Troubleshooting

Problem	Possible Cause	Solution
No Output	Loose Connections	Check all wire connections
Overheating	Overloaded	Check the load
Voltage Fluctuations	Unstable Input	Check the primary voltage source

7. Maintenance

1. Regularly inspect for any signs of wear, tear, or overheating.
2. Keep the transformer clean and free from dust.
3. Tighten any loose connections during routine inspections.

Please read this manual carefully before operating the autotransformer. Failure to follow these instructions may result in damage to the transformer.

9 Cost Analysis

Name of equipment	Cost (Taka)
CRGO EI Lamination Core (1.5kg)	450 (300Tk/kg)
SWG-22 Super Copper Enamel Wire (1.2Kg)	960
Customed Made Bobbin	400
Tapping tube (1meter)	15
Super glue (1piece)	45
Insulating paper (1roll)	30
Masking Tape	40
Transportation	500
Miscellaneous (connectors, iron powder etc)	50
Total	2490

10 Future Work

Better Efficiency: Efficiency is significantly higher than with regular transformers.
Better Voltage regulation: VR outperforms comparable items.
Voltage variability: Voltage output is variable and can change in value as needed.
Huge application sectors:

1. utilized extensively in synchronous and induction motors
2. Due to its assistance in boosting voltage, AC feeders can easily find use for it.
3. In practically all electrical appliances, it is employed.
4. It has long been a part of the electricity transmission and distribution system.

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